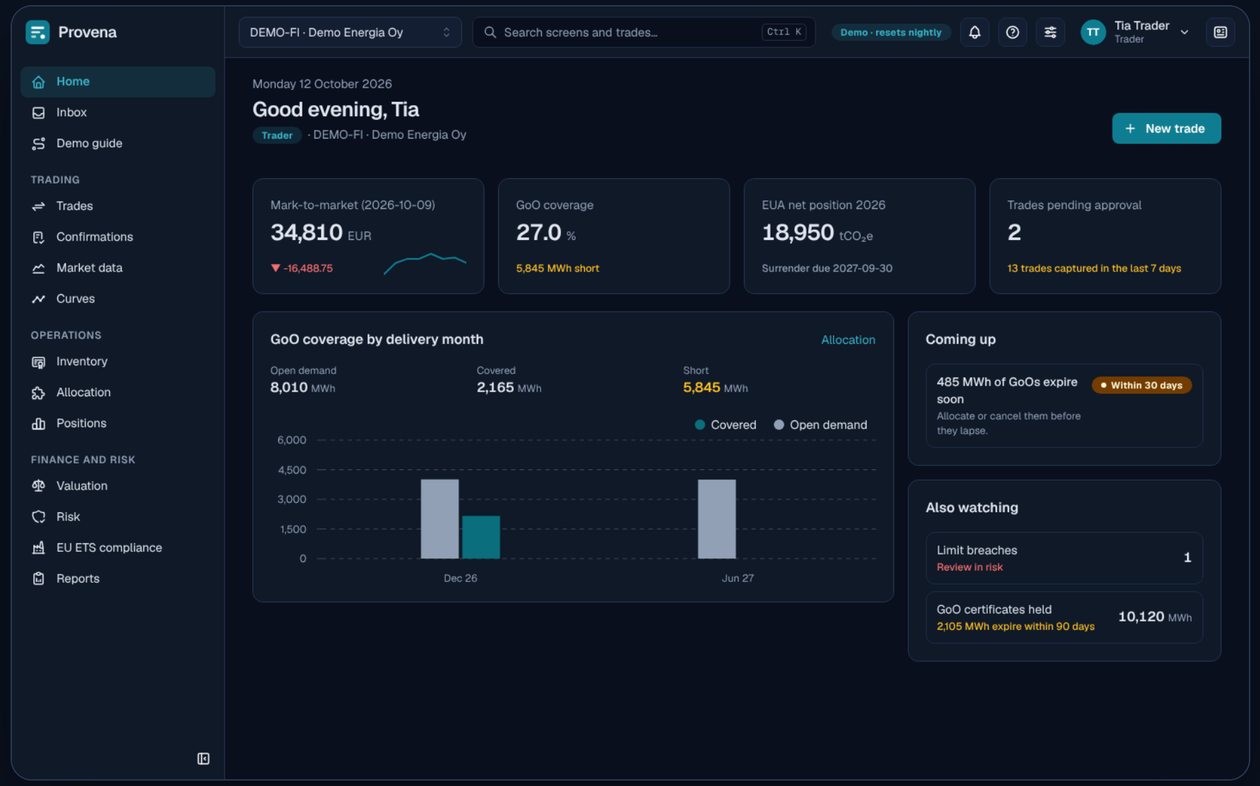
PROVENA

# Certificates and allowances, from trade to ledger

A trading and risk system for guarantees of origin and EU emission allowances, built for audited, regulated operations.



THE PROBLEM

# Why this is hard

## Certificates are ranges

A guarantee of origin is a range of numbers with attributes: country, source, technology, age and support.

## Sales name criteria

A sale says which certificates qualify, not which lots; any lot that matches can deliver.

## Registries differ

The Union Registry, national GoO registries and the AIB hub, through files, APIs and manual steps.

## Two people, fixed dates

Every step that moves value needs a second person, and compliance deadlines are set by law.

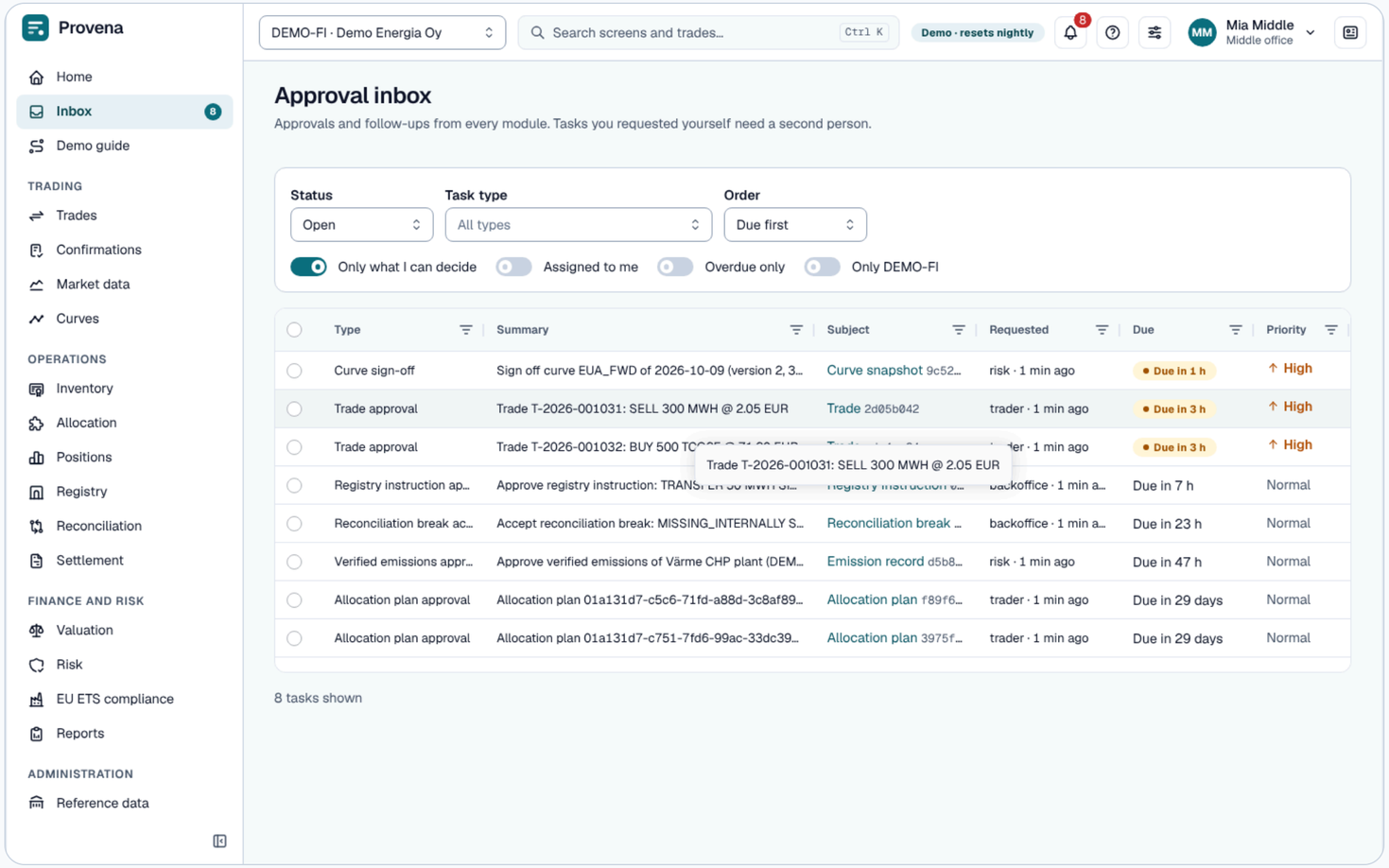
ONE SYSTEM

# Every role, one inbox

Trading, operations, finance, risk and administration in one application.

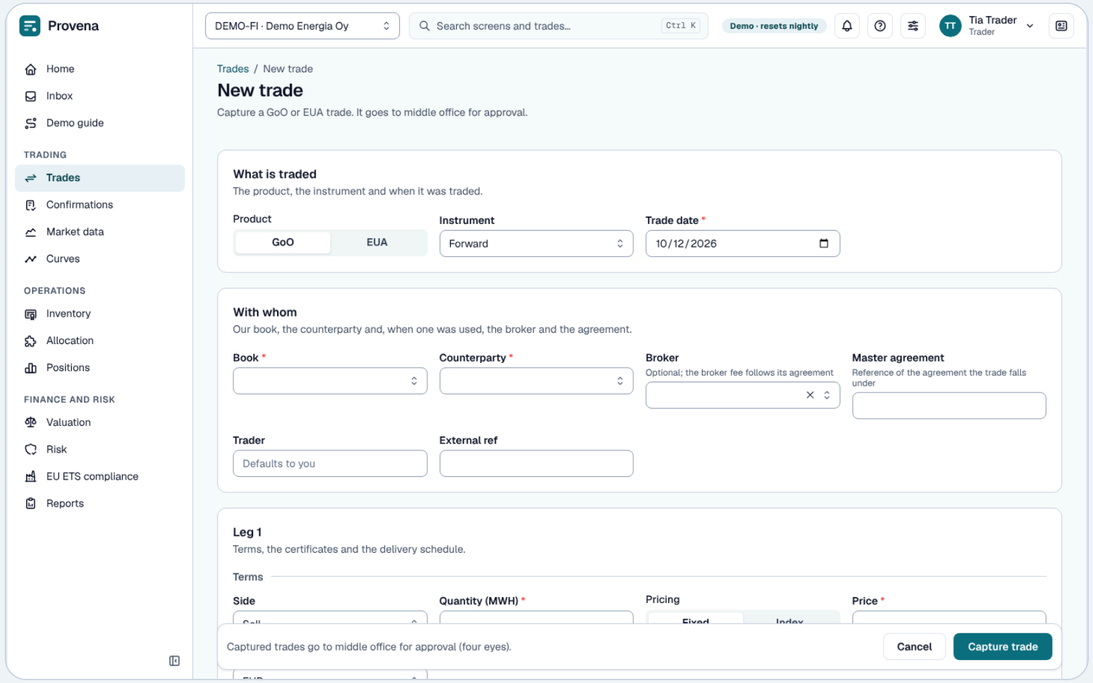
One inbox for every approval, with its four-eyes checks shown.

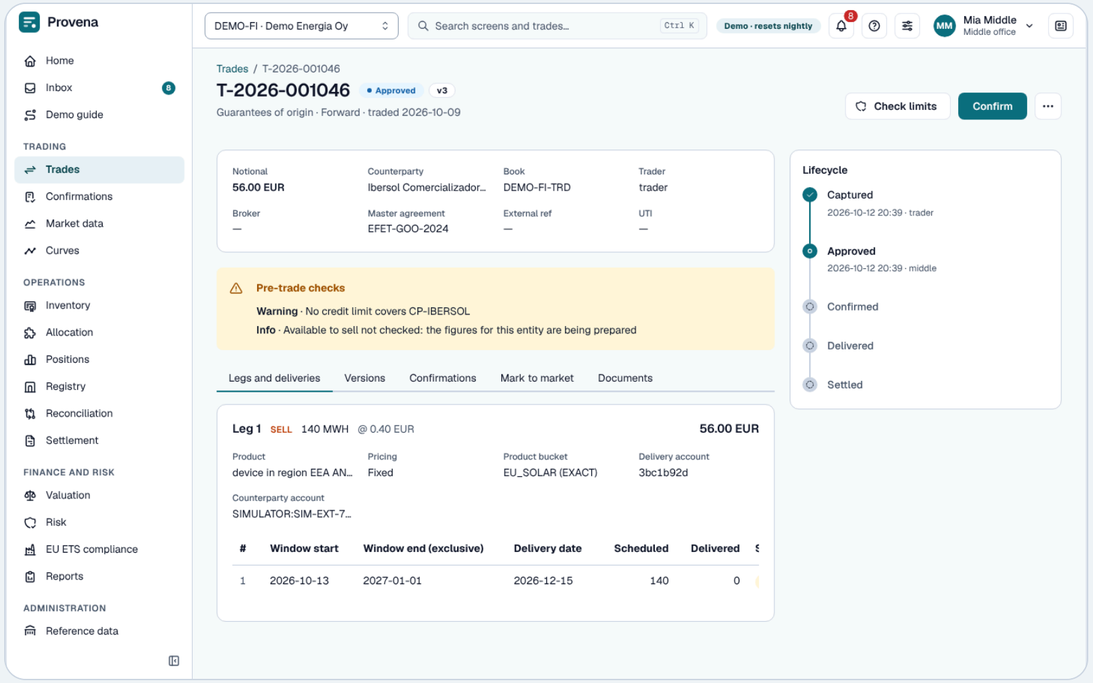
Each role has its own home page and its own work.



TRADING

# Capture, approve, confirm, deliver



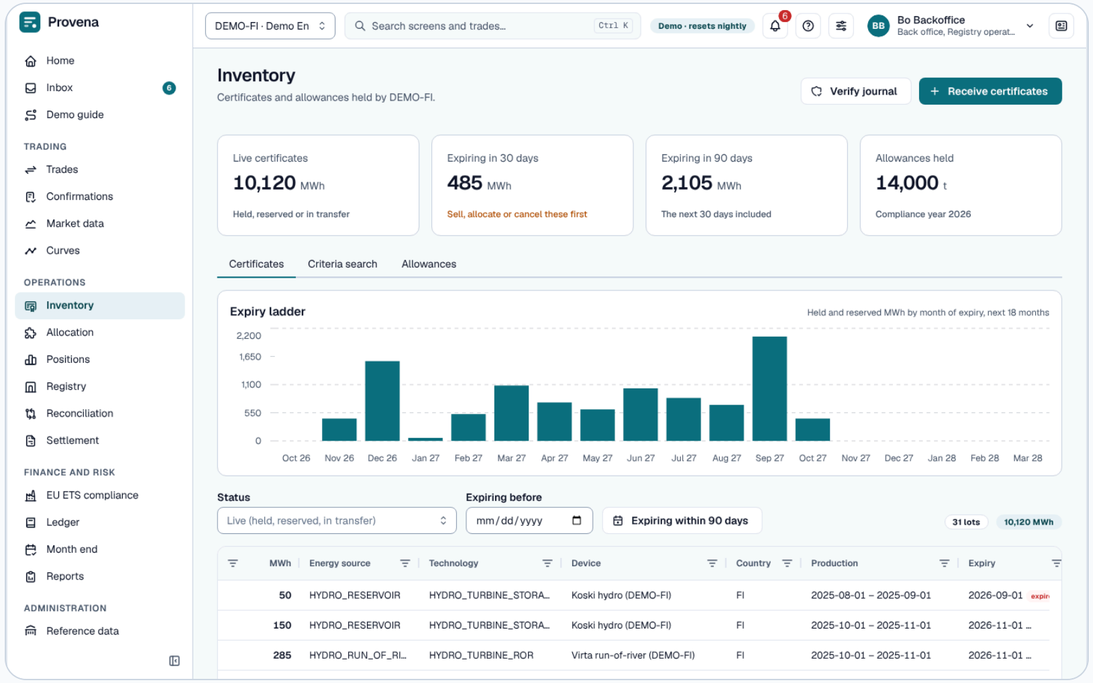

Forwards, spots and futures; fixed or index prices.

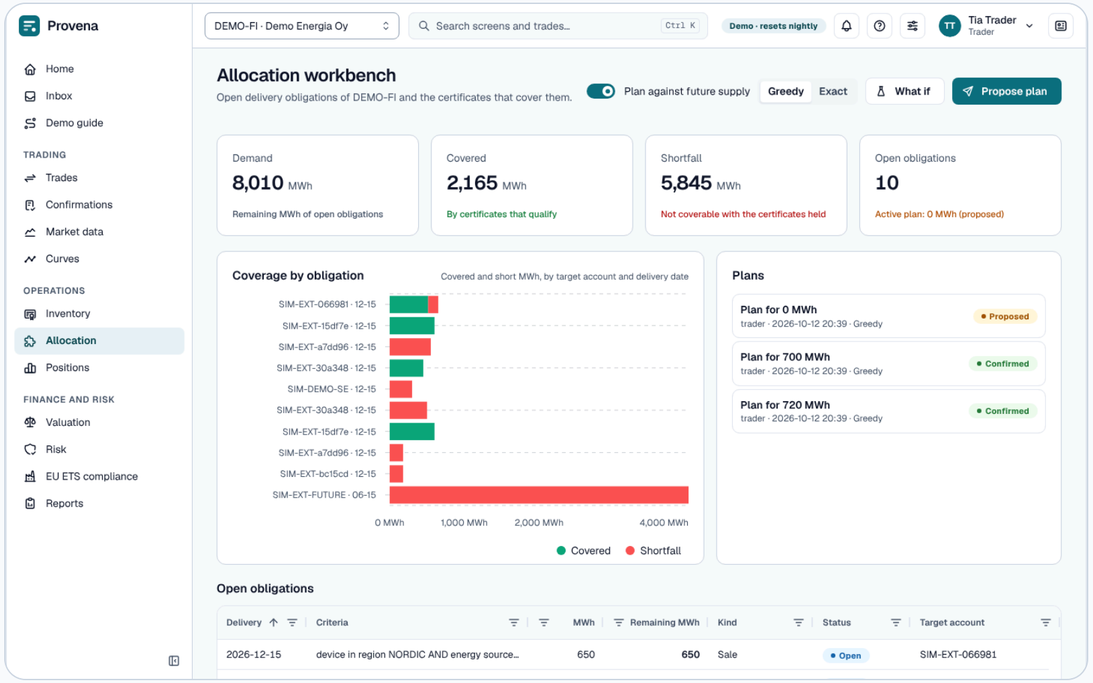
Pre-trade limit checks, then approval by a second person.

Confirmations matched field by field; delivery and settlement.

INVENTORY AND ALLOCATION

# The right certificates for every sale



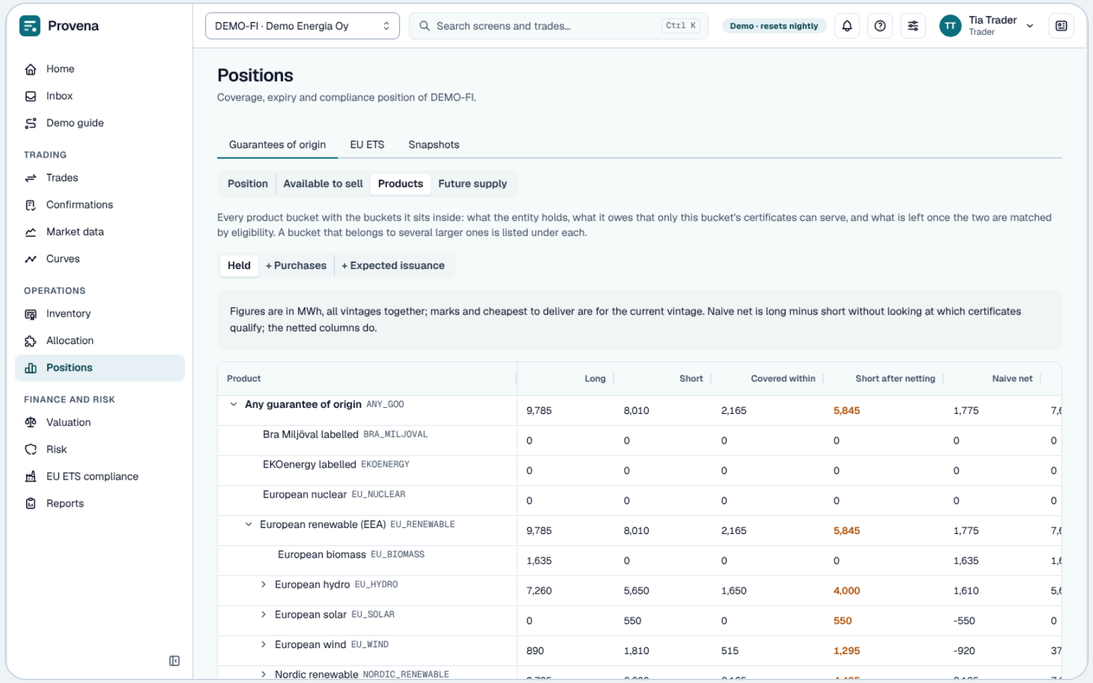

Lots with all their attributes and a hash-chained journal.

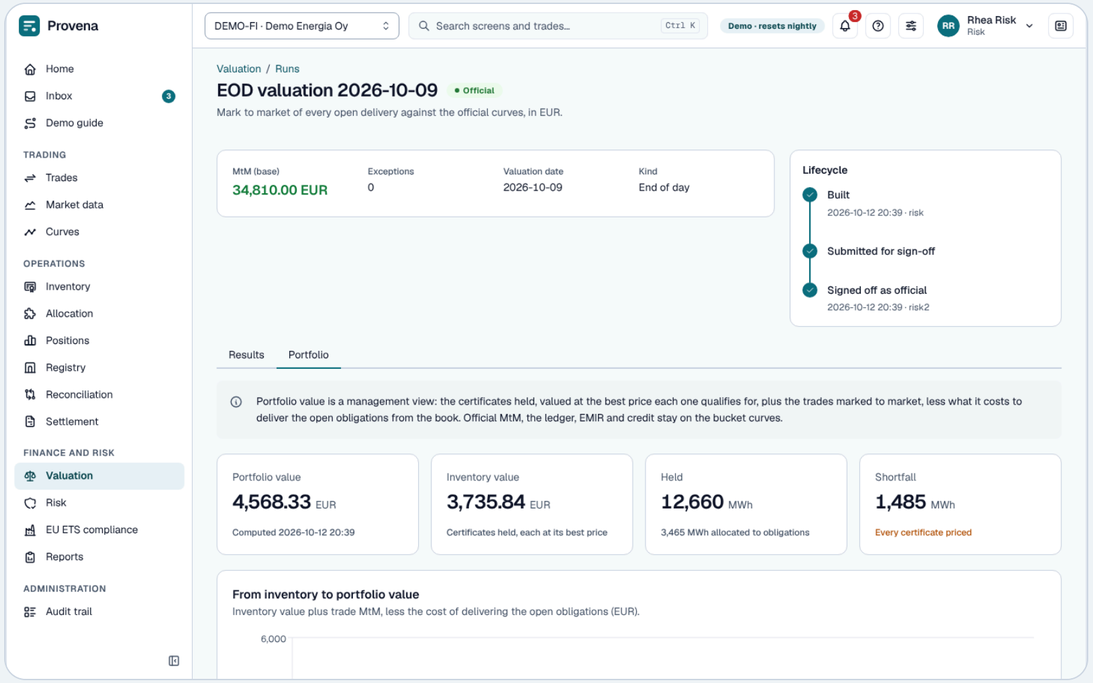
One criteria builder for trades and searches.

Greedy or exact (CP-SAT) allocation, with what-if.

GOO PORTFOLIO MANAGEMENT

# The book as a portfolio, not a list of lots



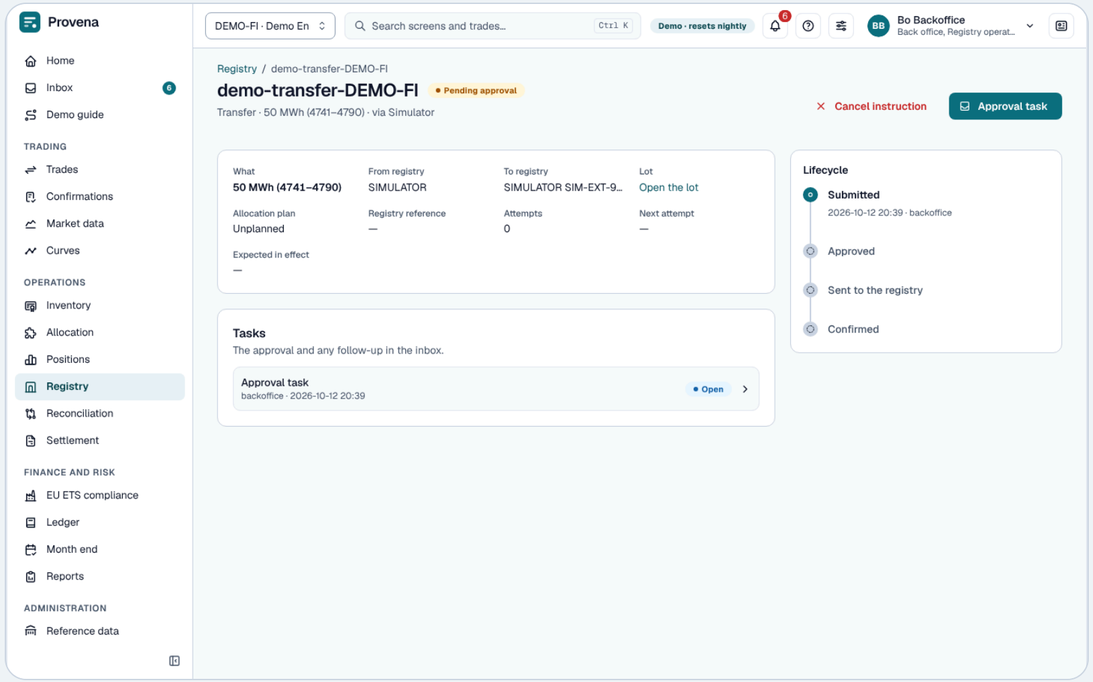

Specific certificates serve generic sales: allocated by value, with the give-away shown.

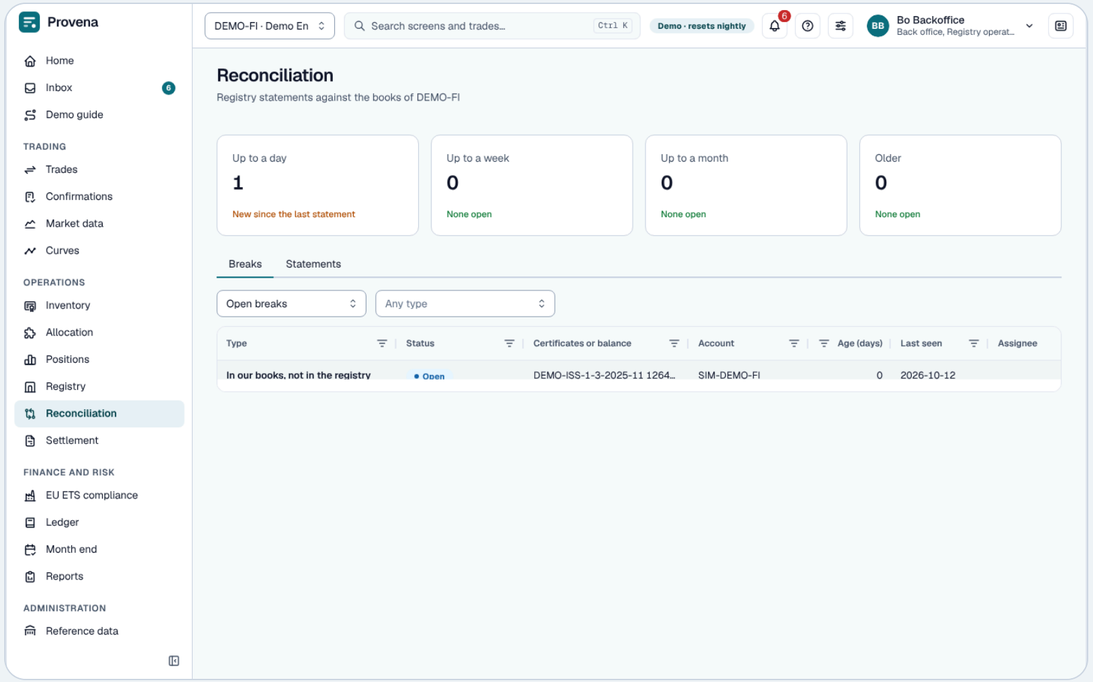
Available to sell, a product tree netted through eligibility, and future supply.

Portfolio value next to each run, and limits on concentration, expiry and basis.

REGISTRY AND RECONCILIATION

# Every transfer accounted for



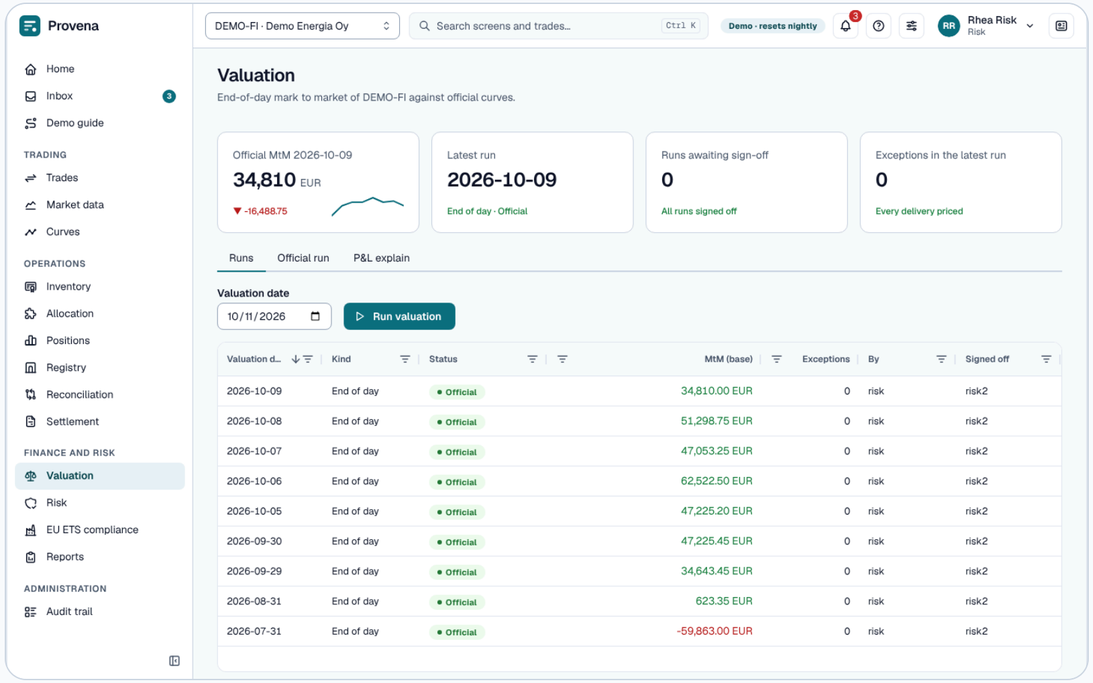

Instructions stored first, sent afterwards; retries are safe.

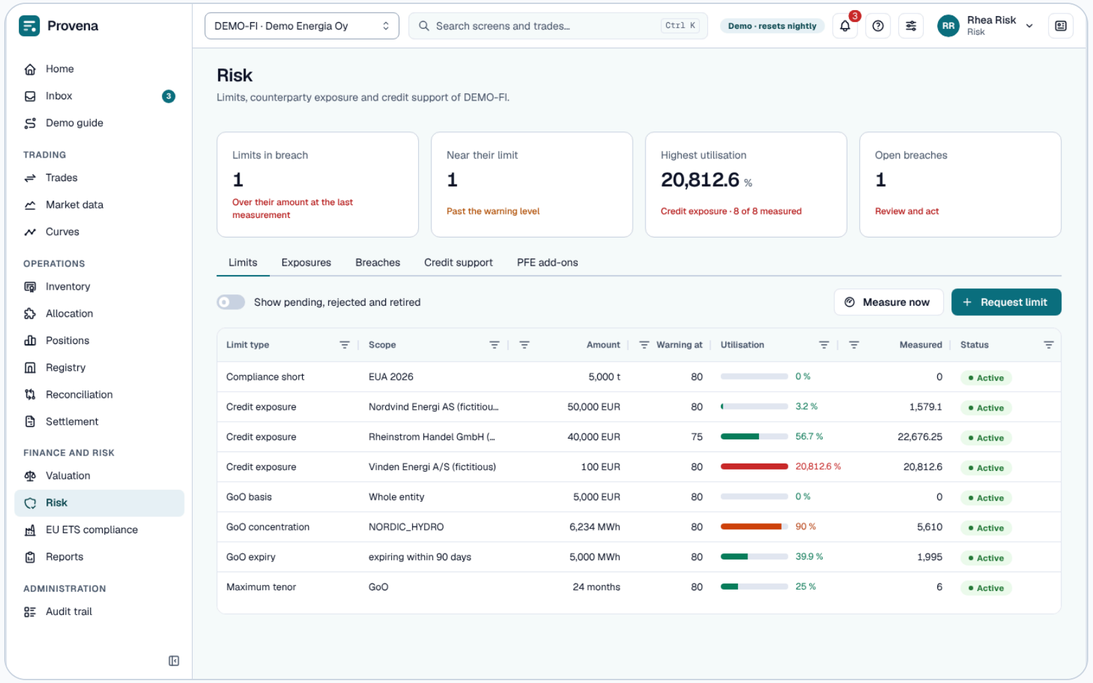
Registry statements reconciled with our books.

Each difference is a break with an owner and an outcome.

MARKET DATA, VALUATION AND RISK

# Prices a second person signs off



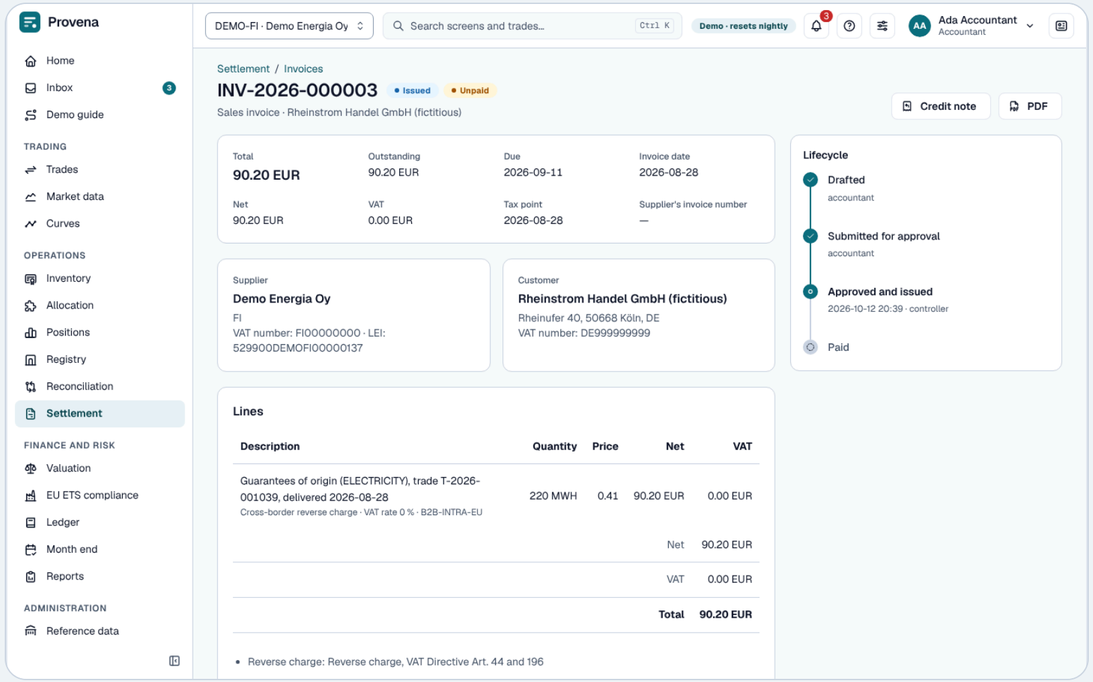

Curves become official once a second person signs them off.

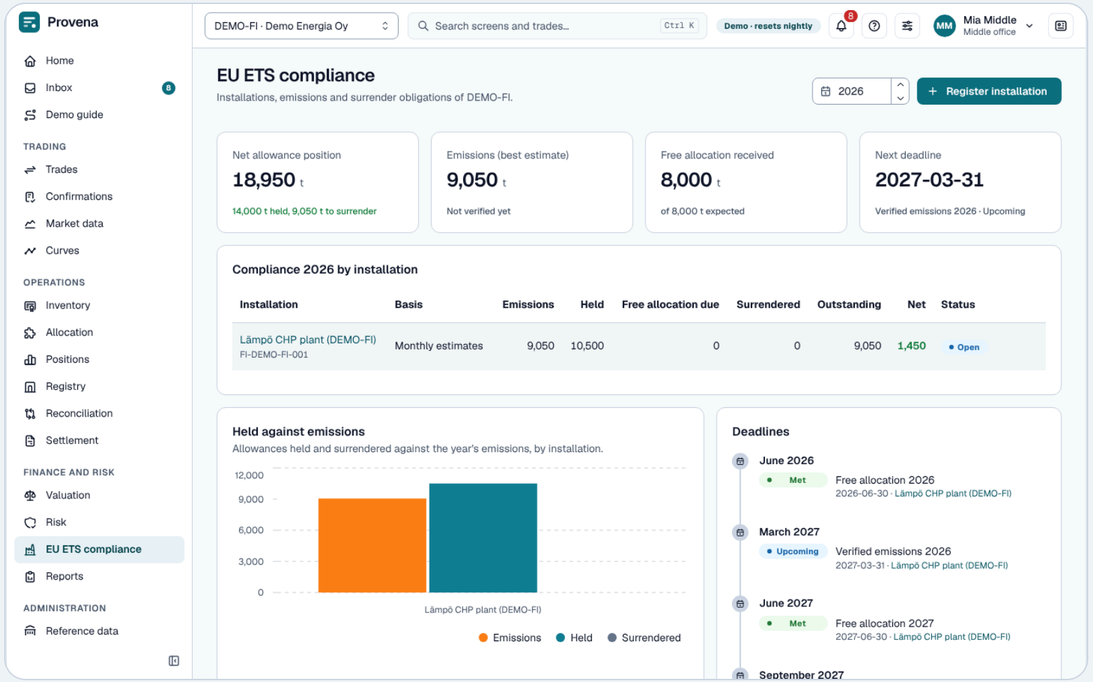
Official end-of-day valuation and P&L explain.

Limits, utilisation, breaches and credit support.

SETTLEMENT AND EU ETS

# Invoices and compliance



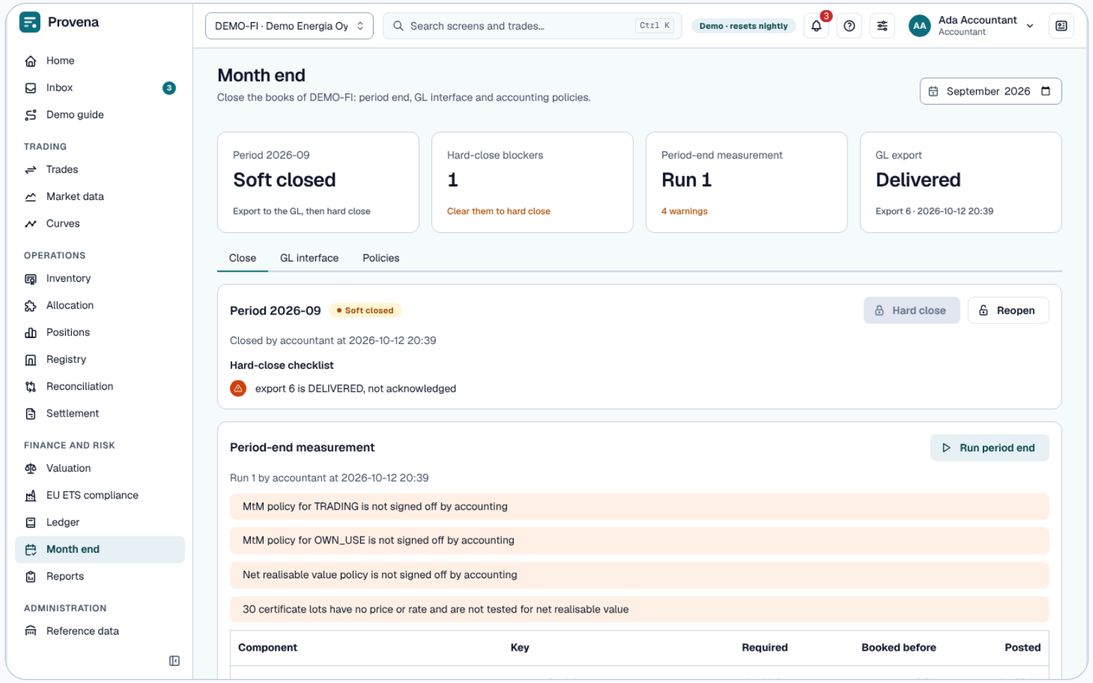

Deliveries become invoices, approved by a second accountant.

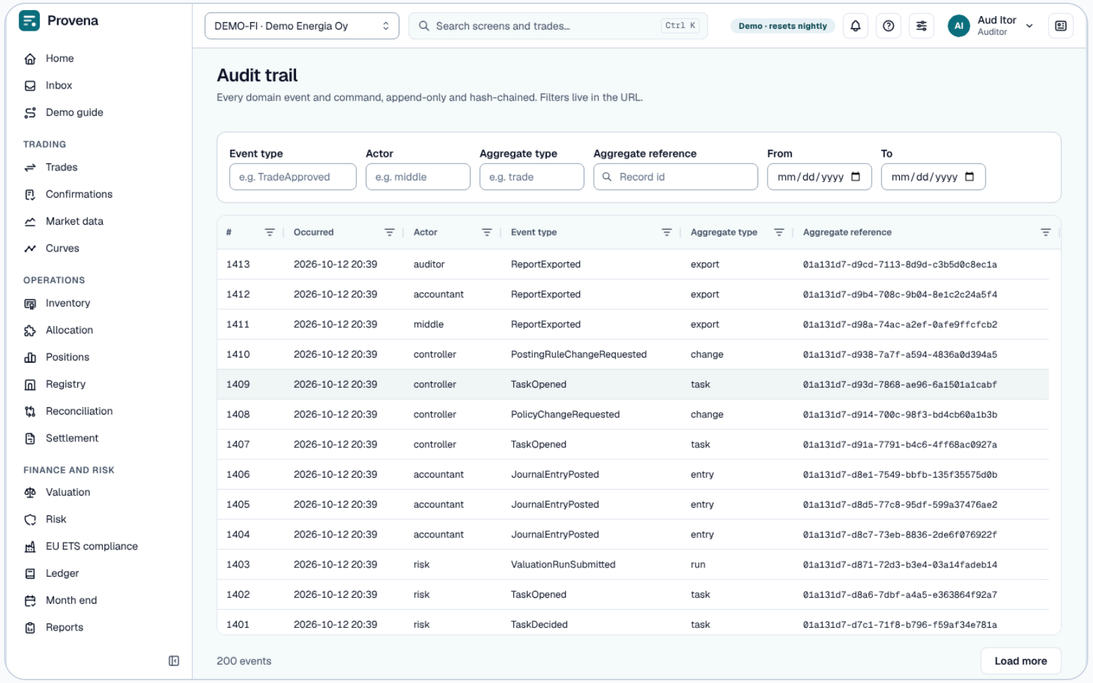
Payments, netting and VAT rules, reverse charge included.

Emissions, free allocation and surrenders per installation.

FINANCE AND CONTROL

# Books an auditor can follow



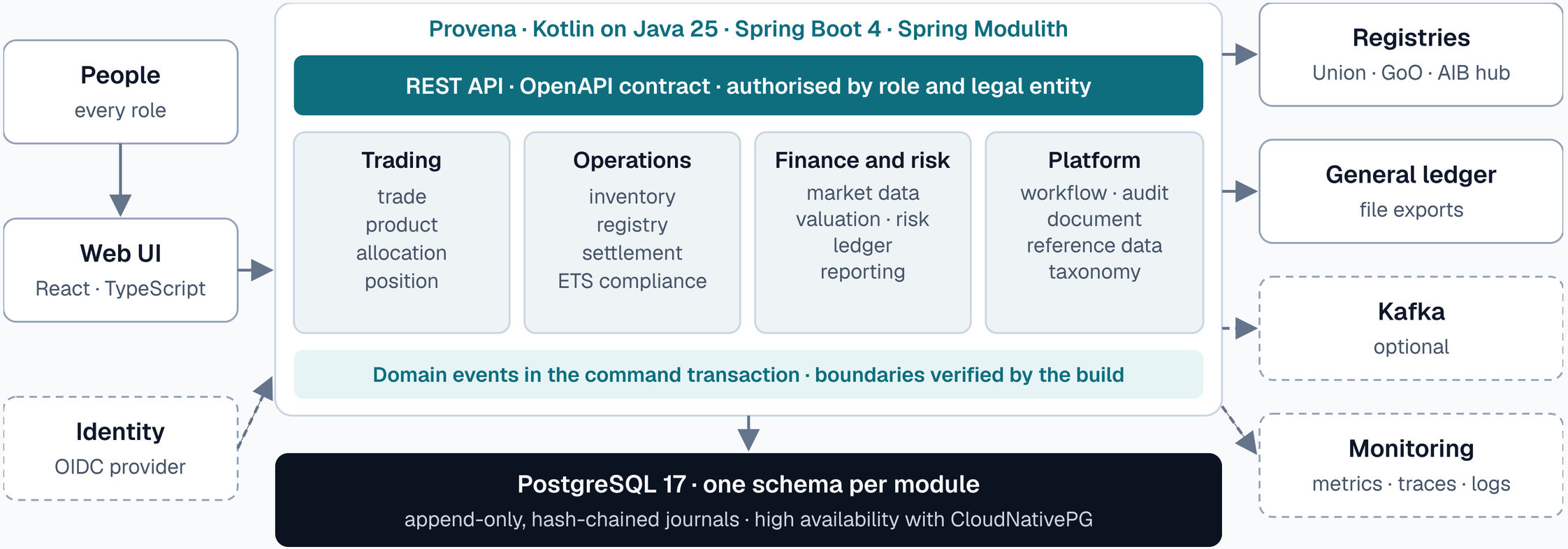

Posting rules feed an append-only, hash-chained journal.

Month end: period end, soft and hard close, GL export.

An audit trail of every change: who, when and why.

ARCHITECTURE

# One application, eighteen bounded contexts



RELIABILITY

# Designed to targets, measured, rehearsed

**99.9 %**
availability target in business hours

**15 min**
recovery point; recovery time 4 hours

**556/s**
requests under stress, 104 users, no errors

**15 ms**
read p95 at baseline load (target 300 ms)

### Stateless, highly available

Horizontal scaling on Kubernetes; PostgreSQL as a three-instance cluster with a synchronous replica.

### Recovery rehearsed

A monthly drill restores a backup, compares row counts and verifies the hash chains.

### Observable

Prometheus metrics, OpenTelemetry traces, structured logs, a trace id on every response.

SECURITY

# Least privilege, two people, a checked supply chain

## Identity and access

Single sign-on with OIDC. Every read and command checks the role and the legal entity it touches.

## Segregation of duties

Four eyes on everything that moves value, enforced in the services and again by the database.

## Hardened web tier

Strict content security policy, CSRF protection, formula-safe exports; ZAP and accessibility checks.

## Supply chain

SBOMs for every build; the image is scanned and a critical fixable flaw stops the release.

ROBUSTNESS AND DATA INTEGRITY

# A system of record that cannot quietly drift

### Append-only, hash-chained

Lot journal, ledger and audit trail are protected by triggers. Errors are reversed, never edited.

### Exactly-once effects

Registry calls through an outbox; idempotent deliveries and GL exports; a timeout is reconciled, not resent.

### Reproducible as of any date

Bitemporal lots, prices and reference data; every report run is logged with its content hash.

### Failures stay contained

A downstream problem is parked for attention; exact decimals only, no floating point in the domain.

MAINTAINABILITY

# Boundaries and contracts the build enforces

## Module boundaries

Each module owns its schema and exposes one API package; any other dependency fails the build.

## Contracts

The OpenAPI document is committed; the web client's types are generated from it, never written by hand.

## Decisions on record

42 architecture decision records, arc42 documentation and a user manual inside the application.

## Strict by default

Warnings are errors; static analysis, a coverage rule, and main moves only on a green pipeline.

ROADMAP

# Nine increments, each one shippable

| Inc. | Delivered |
|---|---|
| 1 | Core domain: products and criteria, trades, inventory, allocation, positions, ledger, audit |
| 2 | Registry integration, reconciliation, settlement and invoicing, confirmations, approval inbox |
| 3 | Market data, curves, valuation, risk and credit, ETS compliance, period end, reporting, GL |
| 4–5 | Web UI for every area, including the operations screens |
| 6 | Operations at scale: observability, supply chain, Kubernetes, HA database, load tests, DR |
| 7 | Showcase: design system, six months of demo data, in-app manual and guided tours |
| 8 | GoO portfolio: value-based allocation, available to sell, future supply, portfolio value, limits |
| 9 | Housekeeping: current dependencies, versioned releases and images, enterprise documentation |

PATH TO PRODUCTION

# What remains needs a partner, not more code

## Assurance

- External penetration test and retest
- DR drill on a production backup
- ETS deadline rules checked by compliance

## Specifications

- Registry APIs (G-REX, NECS) and AIB code lists
- The receiving GL's file layout
- EMIR field layout and trade repository
- Market data vendors

## Sign-offs

- Accounting policies: MtM by book type, NRV, provision
- VAT rules
- PFE add-ons

## Next to build

Faster portfolio refresh on every change, device caps inside the flow, GoO limits per vintage; discounting and hedge accounting; EUA futures margining.

IN NUMBERS, AND TRY IT

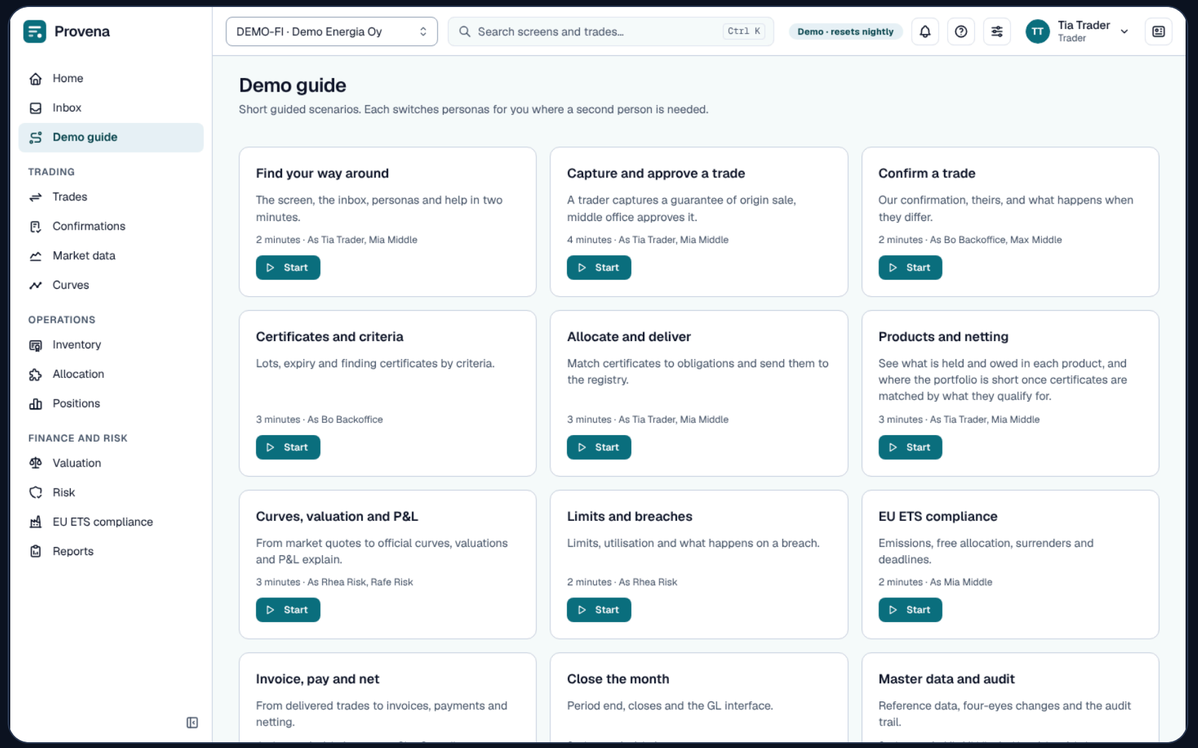
# Evidence you can run yourself

**334**
backend unit and integration tests

**296**
web UI component tests

**82**
browser journeys with accessibility checks

**286 / 303**
API operations with a screen; the rest have a recorded reason




The demo with guided tours and a manual in the app, or your own copy with Docker in minutes.